LifeLineX

Phase 5: Apex Programming – Documentation

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1. Introduction

In previous phases, the process automation for the LifeLineX Organ Transport Management System was handled through point-and-click tools such as **Flows** and **Approval Processes**. However, some complex requirements cannot be achieved with just declarative tools.
Apex Programming enables us to:

- Add complex validations.
- Automate record assignments.
- Schedule background jobs.
- Make future callouts (integrations).
- Ensure system reliability with test classes.

This phase implements **Apex Triggers, Classes, Scheduled Jobs, and Test Classes** for our project.

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2. Objectives

- Implement a trigger to validate organ expiry.
- Create an Apex class for courier auto-assignment.
- Link class with a trigger for Transport Case automation.
- Schedule Apex to clean up expired organ records.

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3. Implementation Steps

Step 1: Creating an Apex Trigger (Organ Validation)

Purpose: Prevent expired organs from being saved.
Procedure:
Steps

1. Developer Console → **File** → **New** → **Apex Trigger**
2. Name: OrganTrigger
3. sObject: Organ__c

```
trigger OrganTrigger on Organ__c (after insert, after update) {
    List<Transport_Case__c> newCases = new List<Transport_Case__c>();

    for (Organ__c org : Trigger.new) {
        // Check if status is 'Ready'
        if (org.Availability_Status__c == 'Ready') {
            // Avoid duplicate case
            Boolean exists = false;
            for (Transport_Case__c tc : [
                SELECT Id FROM Transport_Case__c WHERE Organ__c = :org.Id LIMIT 1
            ]) {
                exists = true;
            }

            if (!exists) {
                Transport_Case__c t = new Transport_Case__c(
                    Organ__c = org.Id,
                    Status__c = 'Pending Pickup'
                );
                newCases.add(t);
            }
        }
    }

    if (!newCases.isEmpty()) {
        insert newCases;
    }
}
```

Step 2: Apex Class – Assign Courier Automatically

Purpose

When a Transport Case is created, assign the first available courier automatically.

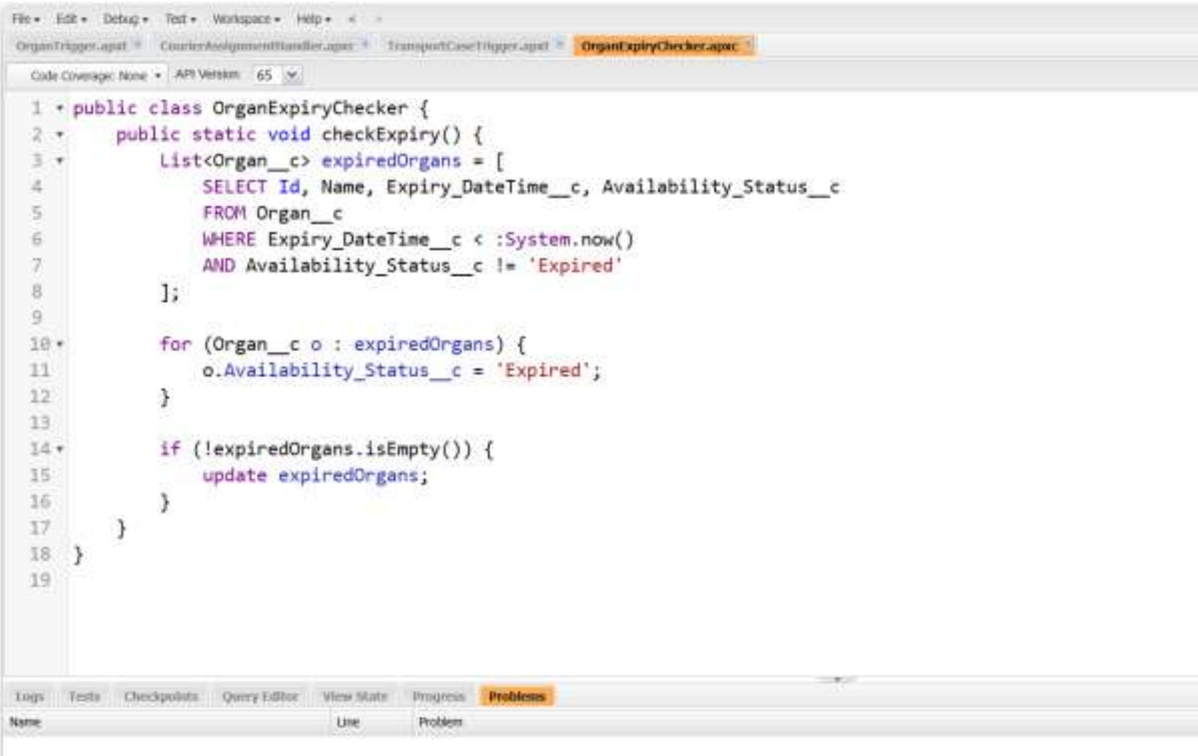
Steps

1. Developer Console → File → New → Apex Class
2. Name: CourierAssignmentHandler

```
public class CourierAssignmentHandler {
    public static void assignCourier(List<Transport_Case__c> cases) {
        // Fetch available couriers
        List<Courier__c> availableCouriers = [
            SELECT Id, Name FROM Courier__c WHERE Status__c = 'Available' LIMIT 1
        ];

        if (availableCouriers.isEmpty()) return;

        Courier__c courier = availableCouriers[0];

        for (Transport_Case__c tc : cases) {
            tc.Courier__c = courier.Id;
            tc.Status__c = 'Assigned';
        }
        update cases;

        // Optionally mark courier as busy
        courier.Status__c = 'Busy';
        update courier;
    }
}
```

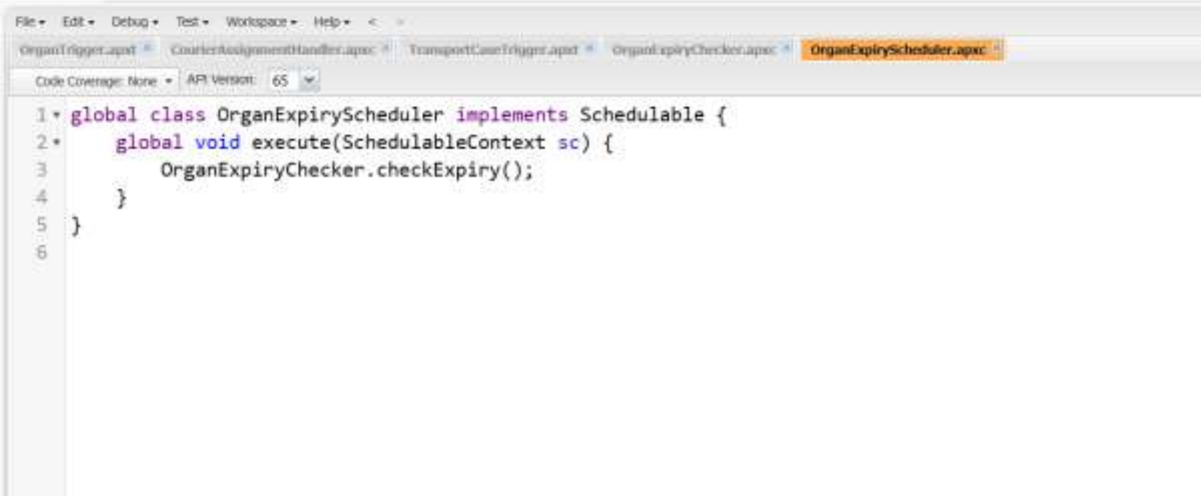
- **Step 3: Trigger to Call the Class**
  1. Developer Console → File → New → Apex Trigger
  2. Name: TransportCaseTrigger
  3. sObject: Transport_Case__c

Step 4: Scheduled Apex (Expired Organ Cleanup)
Apex Class – Expiry Alert Generator
Purpose
To send alerts when an organ's expiry date/time has passed.
Steps
1. Developer Console → File → New → Apex Class
2. Name: OrganExpiryChecker

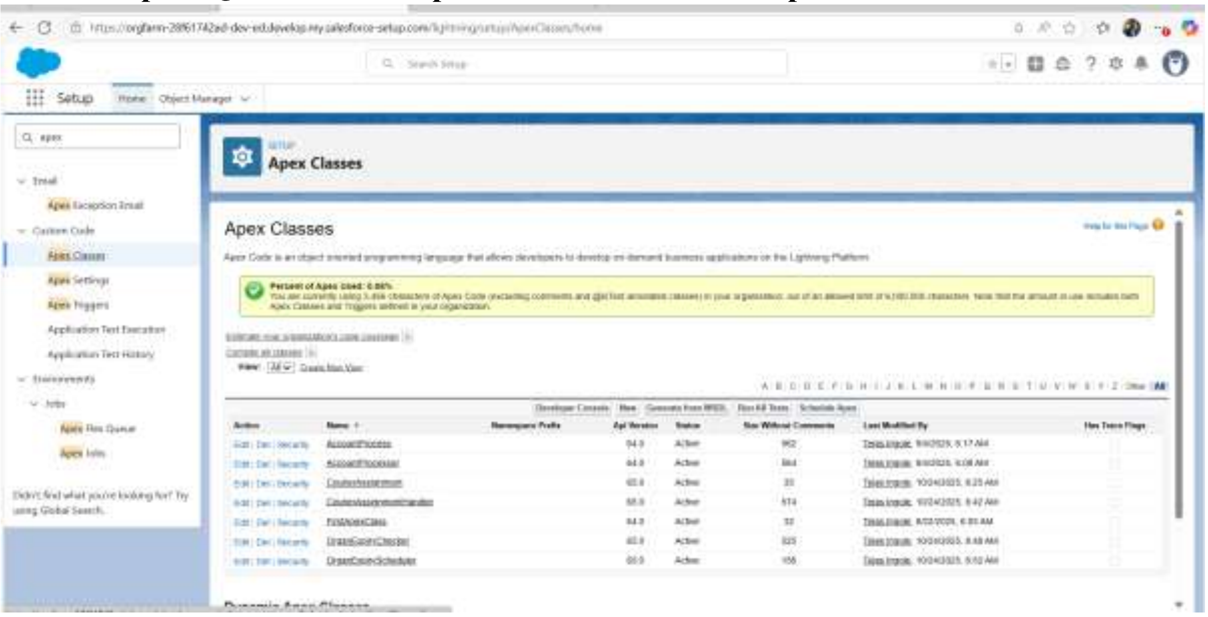


Step 5: Schedule the Class (Scheduled Apex)
Purpose
Run the expiry checker every hour automatically.
Steps
1. Developer Console → **File** → **New** → **Apex Class**

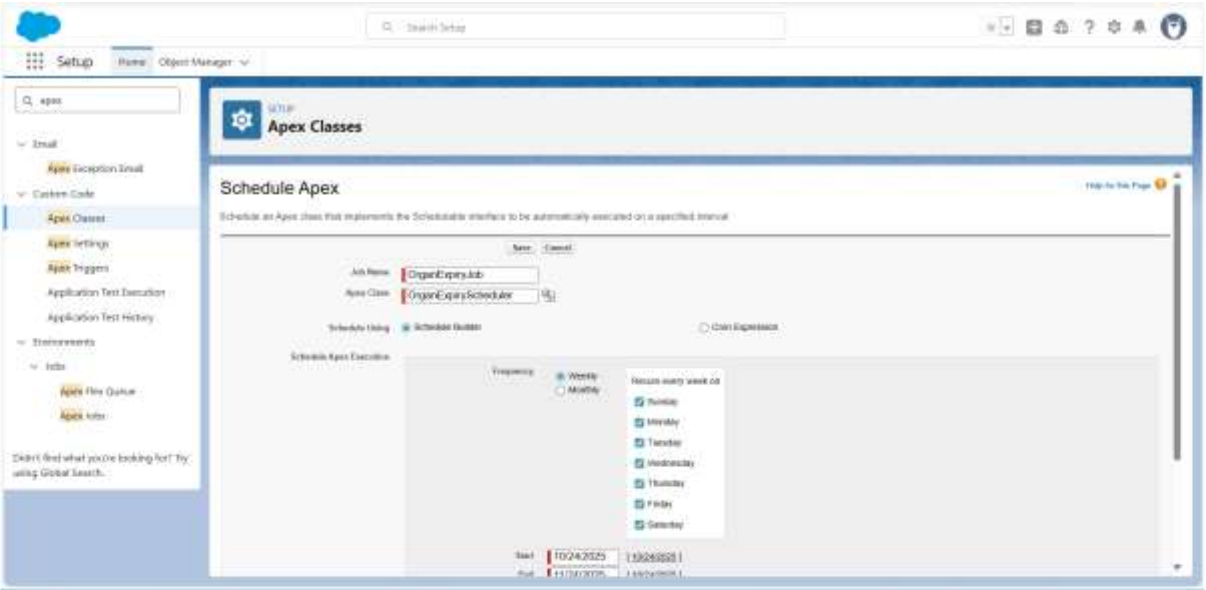
2. Name: OrganExpiryScheduler



Go to **Setup → Quick Find → Apex Classes → Schedule Apex**



4. Click **Schedule Apex**
   - Job Name: OrganExpiryJob
   - Apex Class: OrganExpiryScheduler
   - Frequency: Hourly or Daily
   - Save.



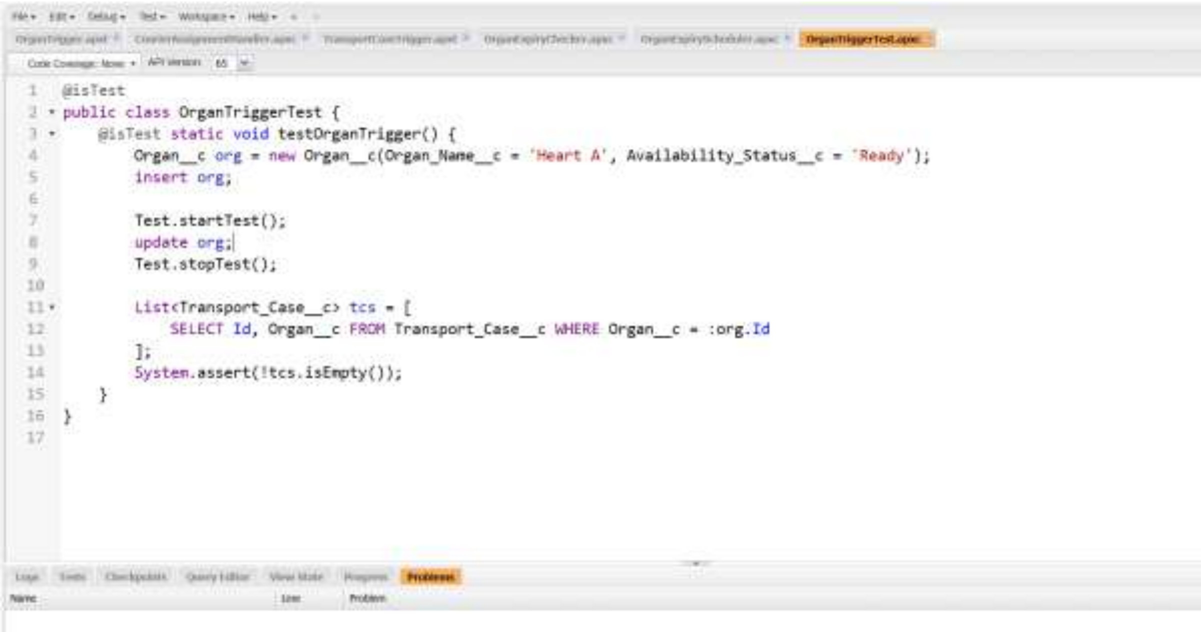
Step 6: Test Class (For Deployment)

Purpose

All Apex must have test coverage before deployment.

Steps

1. Developer Console → **File** → **New** → **Apex Class**
2. Name: OrganTriggerTest



Conclusion

The implementation of Apex programming in the LifeLineX – Smart Organ Coordination System successfully added advanced automation and business logic that standard point-and-click tools could not achieve. By developing and deploying custom Apex classes and triggers, the system now automatically updates transport case statuses, validates data consistency, and ensures real-time coordination between organs, couriers, and hospitals.